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1 SCOPE

1.1 Battery Description

The Li-Ion Volvo battery is Li-Ion energy storage for automotive application (HEV, PHEV, and EV) for energy storage and supply in during buses operation.

According to the preliminary Saft Li-Ion Battery characteristic **Erreur ! Source du renvoi introuvable.** and in order to be compliant with Battery traction specification **Erreur ! Source du renvoi introuvable.**, the Li-ion Battery Architecture which will be taken into account during this analysis could be composed of: 1 Pack including Power Box (BMM) and with ~~Thermal Management its internal cooling circuit.~~

Module: 12 cells in series and 1 SMU
12S of VL30PFfe

Pack: 16 Modules in series with 1 BMM
Pack Type : (16S12S) VL30PFfe
Nominal voltage : 633 VDC (Vmin= 480VDC : Vmax=730VDC)
Typical capacity/energy: ~~31Ah30Ah~~ → ~~22,619~~ kWh

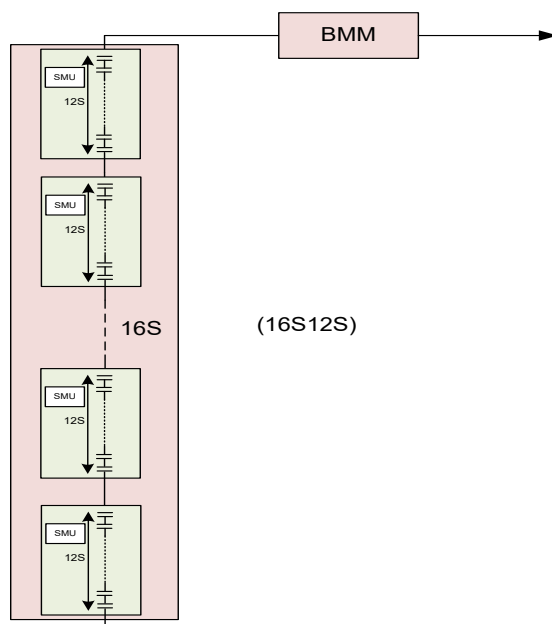


Figure 1: (16S12S) Pack schematic

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1.2 Environment Description (for Battery Application)

1.2.1 Global environment

Depending on the applications, the battery system could be installed:

- On roof of the bus or truck
- Under frame of the bus or truck

According to preliminary functional analysis **Erreur ! Source du renvoi introuvable.**, other equipment could be connected with the Battery:

- On Board charger
- Thermal Management
- Battery 24V
- Loads (Inverter, DC/DC,...)
- Vehicle Mechanical
- Vehicle control system (HPCU)

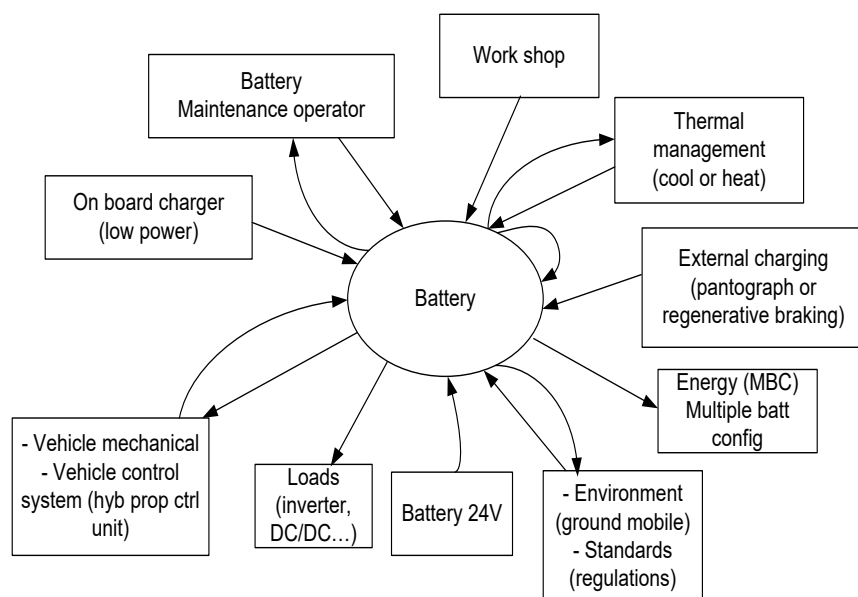


Figure 2: Battery External environments

The batteries could be installed in Energy MBC (Multiple Batteries Configuration).

All application of Li-ion Battery will be for truck or public transportation. In case of a dangerous phenomenon on Battery, the vehicle where the Battery is installed is considered as the hazardous zone.

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1.2.2 Environmental constraints

The Li-ion Battery could be installed on the bus roof, or under frame. The system will operate between -28°C to 80°C (external environment).

1.3 Battery life Cycle

The phases which will be taken into account in the analysis are the following:

- Operating phase,
- Maintenance phase,
- Installation phase,
- Storage phase

The operating phase is composed of the different phases of charge (Fast charge or low power) and discharge of Pack.

Normally at the depot, the vehicle is in power down mode. It means their power supply is switched off. During the maintenance the power supply of the battery has to be switched off.

It is assumed that there is a possibility to charge by regenerative braking or by pantograph and to charge during storage phase. The replacement is possible for Battery. And in case of flood condition, the bus with battery in under frame doesn't operate.

Mis en forme : Couleur de police : Automatique

1.4 Analysis Restriction

The perimeter of this analysis is the battery system mounted on vehicle. The battery system is composed by 1 Pack with its internal cooling-circuit system. All others equipment are not taken into account in this analysis. This analysis is based on VL30PFe Cell behaviors identified at deviation request on SLFP cycling VL30PFe
Erreur ! Source du renvoi introuvable.

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2 ANALYSIS METHODOLOGY

2.1 Hazard and risks analysis methodology

The hazards and risks analysis objectives are:

- To determine the failure function, menace or aggression and the scenario leading to the Hazardous phenomenon of the Li-ion battery system in its application,
- To determine and evaluate the risks related to the Hazardous phenomenon and applicable Hazardous situations, according to their criticality (severity and probability).
- To determine the safety functions and the safety requirements to manage and control the Hazardous event in order to let it tolerable or acceptable, and the other actions which are needed to complete this analysis.

The analysis will take into account the worst case when no safety protection is implemented. The following schematic shows the principles of hazards and risks analysis:

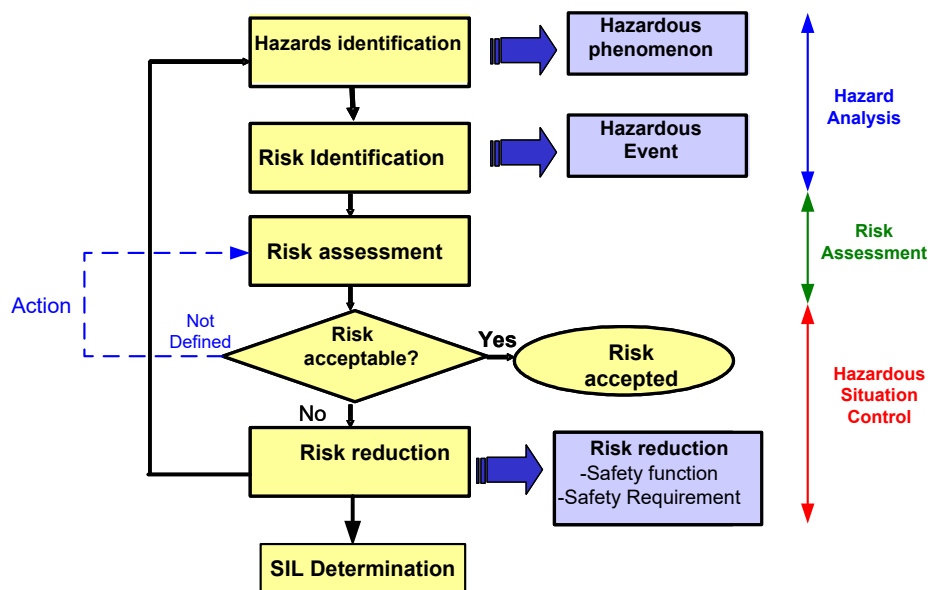


Figure 3: Hazards and risks analysis Methodology

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2.2 Hazard and Risk Analysis Output

The intended outputs from the hazard and risk analysis are:

- Safety function [Cf.**Erreur ! Source du renvoi introuvable.**]
- Safety Requirement [Cf.**Erreur ! Source du renvoi introuvable.**]
- Action [Cf.**Erreur ! Source du renvoi introuvable.**]

2.3 SIL Evaluation Methodology

For each electronic safety function, the risk graph in IEC 61508-5 **Erreur ! Source du renvoi introuvable.**[**DR1**] is used to evaluate its Safety integrity Level.

$$\text{Risque} = C * (F, P, W)$$

The evaluation is based on 4 parameters:

C: Consequence (of the feared event)

F: Frequency of, and exposure time in hazardous zone (Installation concept, packaging ...)

P: Possibility of avoiding the Feared Event (Installation concept, Packaging, Mechanical gas exhaust ...)

W: Probability of unwanted occurrence (without safety function)

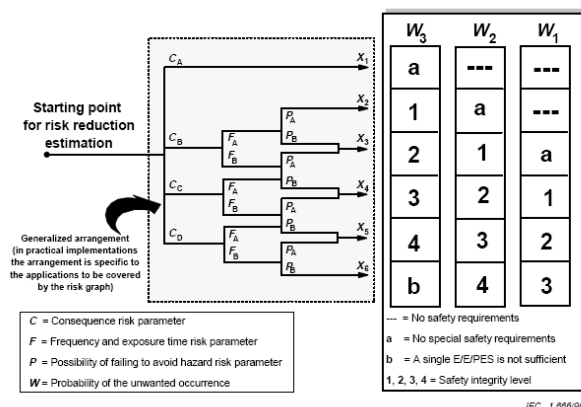


Figure 4: SIL Configuration

The description of C, F, P, W parameters are detailed in Table D.1 of IEC 61508-5 **Erreur ! Source du renvoi introuvable.**[**DR1**] standard. In order to perform the evaluation the unwanted occurrence probability W will be estimated using an interval value (*Table 1*):

Parameter (W)		Classification	Estimation
Probability of unwanted occurrence	W1	A very Slight probability that the unwanted occurrences will come to pass and only a few unwanted occurrences are likely	10^{-9} to 10^{-7} /h
	W2	A slight probability that the unwanted occurrences will come to pass and few unwanted occurrences are likely	10^{-7} to 10^{-5} /h

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Parameter (W)		Classification	Estimation
	W3	A relatively high probability that the unwanted occurrences will come to pass and frequent unwanted occurrences are likely	10^{-5} to $10^{-3}/h$

Table 1: Risk parameter W

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3 HAZARD AND RISK ANALYSIS

In this document, in order to identify the hazardous situation and to reduce their occurrence probability, the following step of Preliminary Hazard Analysis was performed:

- Functional Analysis
- Menace Analysis
- Aggression Analysis

3.1 Menace and Aggression Analysis Perimeter

The following schematics illustrate the differences between both approaches:

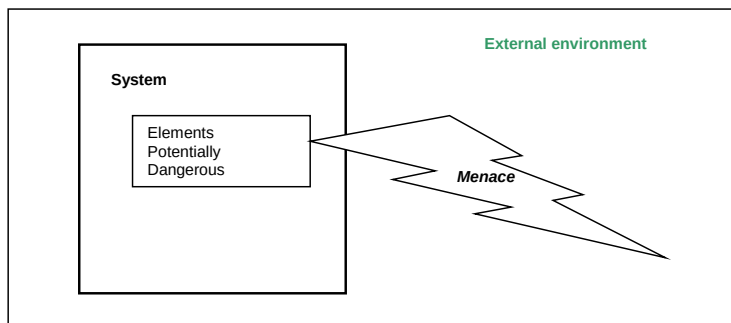


Figure 5: Menace approach

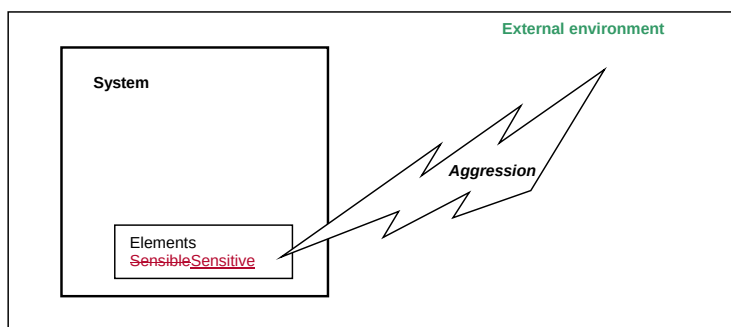


Figure 6: Aggression approach

According to the environmental constraint on the Battery, the following table identifies the eventual hazards which could lead to a feared event. There are two categories of hazard: menaces and aggressions. All identified hazards in this table should be treated during menace and aggression analysis:

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Risk		Identified In					
		System			Environment		
		<i>Menace existing inside the system</i>			<i>Aggression existing in the environment</i>		
Category	Description	Y/N	Dangerous component	Phases concerned	Y/N	Sensible component	Phases concerned
Mechanical	Vibrations						
	Noise						
	Mobile or rotary component						
	Shock, distortion						
	Stress, compression, friction						
	Sharpness, cutting edge						
	Fragility, rip sensitivity						
	Metal projection						
Chemical	Flammable Substance						
	Combustion						

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Risk		Identified In					
		System			Environment		
		<i>Menace existing inside the system</i>			<i>Aggression existing in the environment</i>		
Category	Description	Y/N	Dangerous component	Phases concerned	Y/N	Sensible component	Phases concerned
	Acidic/Basic						
	Corrosive substance						
	Carcinogenic						
	Toxic substance						
	Exploding substances						
	Asphyxiate substances						
Electric	High Voltage						
	High Current						
	Electrostatic energy						
	Electromagnetic radiation						

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Risk		Identified In					
		System			Environment		
		<i>Menace existing inside the system</i>			<i>Aggression existing in the environment</i>		
Category	Description	Y/N	Dangerous component	Phases concerned	Y/N	Sensible component	Phases concerned
	High frequency						
	Presence of ignition source						
Thermodynamic	Source of high temperature (>85°C)						
	Source of cold temperature(<-25°C)						
	Heat transfer source (convection/radiation)						
	High pressure						
	Low pressure						
	Empty						
Radiations	Ionizing substance, radioactive						
	IR, UV						
	Hyper frequency, laser						
Biologic	Virus, bacterium, mushroom						
	Toxin						

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Risk		Identified In					
		System			Environment		
		<i>Menace existing inside the system</i>			<i>Aggression existing in the environment</i>		
Category	Description	Y/N	Dangerous component	Phases concerned	Y/N	Sensible component	Phases concerned
	Vermin						
Environmental	Humidity, condensation						
	Flood						
	Ice						
	Rain						
	Snow						
	Lightning						
	Wind						
	Hailstorm						
	Salt						
	Sand, dust						

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Table 2: List of constraints for menace and aggression analysis

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All dangerous components from Battery are considered as menace for external environment and will be analyzed during menace PHA. All Hazardous phenomena from environment are considered as aggression for Battery and will be analyzed during aggression PHA.

3.2 Hazard and Risk analysis formalism

The formalism used for this analysis is the following:

3.2.1 Functional PHA

- [A] : "Function", description of function which is performed by the intrinsic component of Battery (It is available only during Functional PHA)
- [B] : "Component", what compose the function in column [A] Component of a function leading to dangerous phenomenon (It is available only during Functional PHA)
- [C] : "Failure function/component" Kind of failure which could happen on each function (It is available only during Functional PHA)
- [D] : "Causes", the causes which are the origin of the failure mode, Kind of failure which could happen on each function (It is available only during Functional PHA)
- [E] : "Phase", each failure is identified and analysed for each step of operational phase.
- [F] : "Scenario", the effect of the failure on the system
- [G] : "Dangerous Phenomenon", Phenomenon which is considered as dangerous
- [H] : "Dangerous Situation": Situation in which a person or material is exposed to dangerous phenomenon
- [I] : "Feared Event" is the result of a dangerous phenomenon and dangerous situation.
- [J] : "Probability Level" → "P" (cf. appendix 4.1.1)
- [K] : "Severity Level" → "S" (cf. appendix 4.1.2)
- [L] : "Risk Assessment" (cf. appendix 4.1.3)
- [M] : "Possibility of avoiding the risk" which safety requirement or safety function must be implemented to reduce the probability or the severity of the event.
- [N] : Re-evaluation of Probability with possibility of avoiding the risk
- [O] : Re-evaluation of Severity with possibility of avoiding the risk
- [P] : Re-evaluation of risk with possibility of avoiding the risk

The aim of the last column [M] in the table is to suggest safety functions or safety requirements in order to consider the risk assessment value as tolerable or acceptable (instead of intolerable).

The approach of the safety studies primarily focuses on (by order of preference):

- Elimination of hazards
- Decrease gravity

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Mis en forme : Police :Non Gras

Mis en forme : Police :Non Gras, Anglais (États-Unis)

Mis en forme : Police :Non Gras

Mis en forme : Anglais (États-Unis)

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- Decrease the probability of occurrence

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Function	Component	Failure Function/ component	Causes	Phase	Scenario	Hazardous phenomenon	Hazardous Situation	Hazardous event	Proba	Severity	Risk Assessm ent	Possibility of avoiding the risk	Re-evaluation Proba	Re-evaluation Severity	Re-evaluation Risk
[A]	[B]	[C]	[D]	[E]	[F]	[G]	[H]	[I]	[J]	[K]	[L]	[M]	[N]	[O]	[P]

Table 3: Functional PHA formalism

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3.2.2 Menace PHA

- [A] : "Risk Category", the category of risks which is considered as a menace and which corresponds to the hazard
- [B] : "Dangerous component", Component which can lead to a menace against the external environment and corresponding to the risk category of the first column
- [C] : "Phase" the phase in which the risk is considered.
- [D] : "Risk Causes and origins", the causes which are the origin of the risk.
- [E] : "Phenomena explanation", explanation of the system behaviour in presence of the hazard.
- [F] : "Dangerous Phenomenon", Phenomenon which is considered as dangerous
- [G] : "Dangerous Situation", Situation in which a person or material is exposed to dangerous phenomenon
- [H] : "Feared Event" is the result of a dangerous phenomenon and dangerous situation.
- [I] : "Probability Level" → "P" (cf. appendix 4.1.1)
- [J] : "Severity Level" → "S" (cf. appendix 4.1.2)
- [K] : "Risk Assessment value" (cf. appendix 4.1.3)
- [L] : "Possibility of avoiding the risk" which safety requirement or safety function must be implemented to reduce the probability or the severity of the feared event.
- [M] : Re-evaluation of Probability with possibility of avoiding the risk
- [N] : Re-evaluation of Severity with possibility of avoiding the risk
- [O] : Re-evaluation of Risk with possibility of avoiding the risk

3.2.3 Aggression PHA

- [A'] : "Risk Category", the risk category which is considered as an aggression
- [B'] : "Sensitive component", Component which can be impacted by the aggression from the external environment (It is available only during Aggression PHA). The detail of the aggression is considered in the first column
- [C'] : "Phase"
- [D'] : "Risk Causes and origins"
- [E'] : "Phenomena explanation"
- [F'] : "Dangerous Phenomenon"
- [G'] : "Dangerous Situation"
- [H'] : "Feared Event"
- [I'] : "Probability Level" → "P"
- [J'] : "Severity Level" → "S"
- [K'] : "Risk Assessment value"

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- [L'] : "Possibility of avoiding the risk"
[M'] : Re-evaluation of Probability with possibility of avoiding the risk
[N'] : Re-evaluation of Severity with possibility of avoiding the risk
[O'] : Re-evaluation of risk with possibility of avoiding the risk

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Risk Category	Dangerous Component (menace)	Phase	Risk causes and origins	Phenomena explanation	Dangerous phenomena	Dangerous situation	Feared Event	P	S	Risk assessment value	Possibility of avoiding the risk	Re-evaluation P	Re-evaluation S	Re-evaluation Risk
[A]	[B]	[C]	[D]	[E]	[F]	[G]	[H]	[I]	[J]	[K]	[L]	[M]	[N]	[O]

Table 4: Menace PHA formalism

Risk Category (aggression)	Sensitive component	Phase	Risk causes and origins	Phenomena Explanation	Dangerous Phenomena	Dangerous Situation	Feared Event	P	S	Risk assessment value	Possibility of avoiding the risk	Re-evaluation P	Re-evaluation S	Re-evaluation Risk
[A']	[B']	[C']	[D']	[E']	[F']	[G']	[H']	[I']	[J']	[K']	[L']	[M']	[N']	[O']

Table 5: Aggression PHA formalism

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Li-ion Volvo Battery****4 APPENDIX****4.1 APPENDIX B: RISK SCALE****4.1.1 Probability scale**

Using the probability table below, the hazardous situation occurrence probability (occurrence Probability of the Feared event) is evaluated. The frequencies of hazard cases are defined as:

N°	Category	Description
6	Frequent $P > 10^{-3}/h$	Likely to occur frequently. The hazard will be continually experienced
5	Probable $10^{-5}/h < P \leq 10^{-3}/h$	Will occur several times. The hazard can be expected to occur often
4	Occasional $10^{-7}/h < P \leq 10^{-5}/h$	Likely to occur several times. The hazard can be expected to occur several times
3	Remote $10^{-9}/h < P \leq 10^{-7}/h$	Likely to occur sometime in the system life cycle. The hazard can reasonably expected to occur
2	Improbable $10^{-11}/h < P \leq 10^{-9}/h$	Unlikely to occur but possible. It can be assumed that the hazard may exceptionally occur.
1	Incredible $P \leq 10^{-11}/h$	Extremely unlikely to occur. It can be assumed that the hazard may not occur.

4.1.2 Severity scale

The table below is used to evaluate the final effect on user of a hazardous event due to battery failure. The hazard levels ~~have are~~ the following:

N°	Severity Level	Consequence to Persons or Environment	Consequence to Service
4	catastrophic	Fatalities and/or multiple severe injuries and/or major damage to the environment	
3	critical	Single fatality and/or severe injury and/or significant damage to the environment.	Loss of a major system
2	marginal	Minor injury and/or significant threat to the environment	Severe system(s) damage
1	insignificant	Possible minor injury	Minor system damage

4.1.3 Risk assessment matrix

A risk level is the combination of Hazard severity and Occurrence probability. The approach of the safety studies primarily focuses on hazards and retains elimination or severity reduction of the hazards as a preferred method. Final overall risk quantification is then performed to verify the compliance to the safety target.

However, during the safety studies, a categorisation for individual hazardous scenario may be considered where Risk Reduction is performed. In this case, the following rankings will be considered:

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Level	Frequency of occurrence of a hazardous event	Risk Levels			
6	frequent	undesirable	intolerable	intolerable	intolerable
5	probable	tolerable	undesirable	intolerable	intolerable
4	occasional	tolerable	undesirable	undesirable	intolerable
3	remote	negligible	tolerable	undesirable	undesirable
2	improbable	negligible	negligible	tolerable	tolerable
1	incredible	negligible	negligible	negligible	negligible
		insignificant	marginal	critical	catastrophic
		1	2	3	4
		Severity Levels of Hazard Consequence			

The qualitative risk categories of the hazard evaluation are defined:

Risk Evaluation	Risk reduction/control
Intolerable	Shall be eliminated
Undesirable	Shall only be accepted when risk reduction is impracticable and with the customer agreement
Tolerable	Acceptable with customer agreement
Negligible	Acceptable without any agreement

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